CASH LOGISTICS

Right cash, right place

Forecast-driven cash logistics for the Khon Kaen region · MVP on Databricks

–8% CIT cost/day | **Cash-out risk 64 → 0** | **100% on-time**

CURRENT STATE

Where we are today

159
branches

520
machines: 400 ATMs and 120 cash recyclers

3
cash centres

30
armoured vans

5-day
fixed replenishment cycle, whatever the demand

MVP SCOPE

One region
Khon Kaen first

One decision
Where cash goes tomorrow

Two goals
Less idle cash · zero cash-outs

THE PAIN, IN NUMBERS · PLAN DAY 1 OCT 2026

What it costs today

IDLE CASH

~THB 1.56B

sitting in machines and vaults on the plan day

ON-TIME SLA

14.7%

of stops inside branch opening hours

CASH-OUT RISK

64

machines at risk of running dry in one day

INEFFICIENCY

135 stops

and 1,711 km for demand that needs only **97 stops**

> A calendar, not demand, decides where cash goes: vans visit full machines while busy ones run dry.

Opportunity cost (illustrative): at an assumed 2% cost of funds, every THB 43M of float freed ≈ THB 0.9M a year.

WHY NOW

Why now

TO CONFIRM WITH YOU

Cash usage is falling

PromptPay and QR payments keep growing, so a fixed cycle overstocks machines more every year.

TO CONFIRM WITH YOU

Peaks ahead

New Year and Songkran payday peaks are exactly when a calendar rule fails and machines run dry.

TO CONFIRM WITH YOU

Cost pressure

CIT and fuel costs are rising, and the vendor contract and budget cycle create a natural decision point.

TO CONFIRM WITH YOU

Cost of waiting

≈ THB 125K a month in CIT cost alone, before idle cash and cash-outs.

ROOT CAUSES

Why it happens

1 · FRAGMENTED DATA

ATM switch, vault system, CIT trip logs, core banking and holiday spreadsheets live apart.

2 · KNOW-HOW STUCK WITH PEOPLE

Drivers and dispatchers know which sites are closed or busy; it travels by radio.

3 · NO FORECAST

A calendar rule decides the visit, not expected withdrawals.

4 · DAY-OLD MANUAL PLANNING

Routes are built in Excel from yesterday's numbers.

5 · NO FEEDBACK LOOP OR AUDIT

Overrides happen by phone; nobody can see why a stop changed.

6 · NO COMMON SECURITY MODEL

Custodian PII and depot access are controlled differently in every system.

> **“Data, forecast and people's know-how never meet in one place.”**

THE SOLUTION

The solution in four steps

01

UNIFY

Lakeflow pipelines + Unity Catalog

fixes: fragmented data (1), no common security model (6)

02

PREDICT

Forecasting model on Model Serving

fixes: no forecast (3)

03

DECIDE

Route optimizer (OR-Tools)

fixes: day-old manual planning (4)

04

ACT

Databricks App + Lakebase + Genie

fixes: know-how stuck with people (2), no feedback loop or audit (5)

Unity Catalog governance, end to end Tags · PII mask · Depot row filter · Lineage raw file → app

Each step reads the previous step's governed output: one copy of the data, one security model, from raw feed to the dispatcher's screen.

LIVE DEMO · PLAN DAY 1 OCT 2026

What I'll show you

1	**Overview** Health map and KPI cards for the region	→	ANSWERS **Current state & idle cash**
2	**Route Tracking** Baseline (fixed cycle) vs optimized plan	→	ANSWERS **Inefficiency & on-time SLA**
3	**Machines · re-forecast** 7-day forecast for one machine, live from the serving endpoint	→	ANSWERS **Cash-outs**
4	**Truck Detail · override** Skip a stop with a reason; saved to Lakebase	→	ANSWERS **Driver know-how & audit**
+	**Optional: Ask Cash Ops (Genie)** Plain-language questions over the same governed tables	→	ANSWERS **Self-service answers, no new reports**

Backup: plan vs today

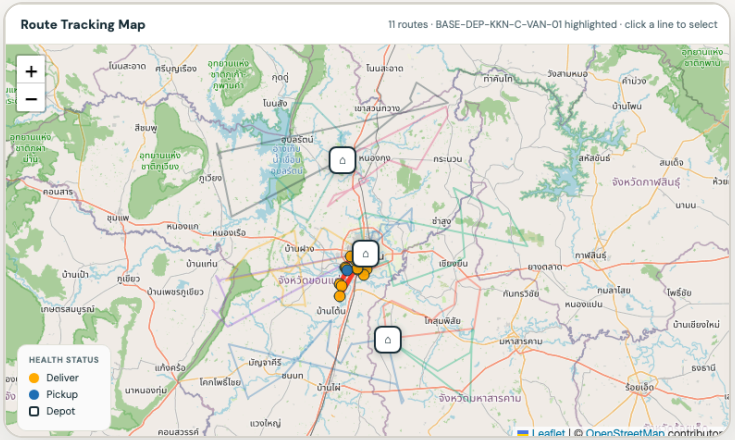
TODAY · FIXED 5-DAY CYCLE

CIT COST / DAY	DISTANCE	VANS USED	IDLE CASH	CASH-OUT RISK NODES	SLA	CREW HOURS
฿75K	1,711 km	11 / 30	฿1561.32M	64	14.7%	92 h
▲ 8.4% vs optimized	▲ 29.3% vs optimized	▼ 15.4% vs optimized	▲ 2.8% vs optimized		▼ 85.3% vs optimized	▲ 49.8% vs optimized

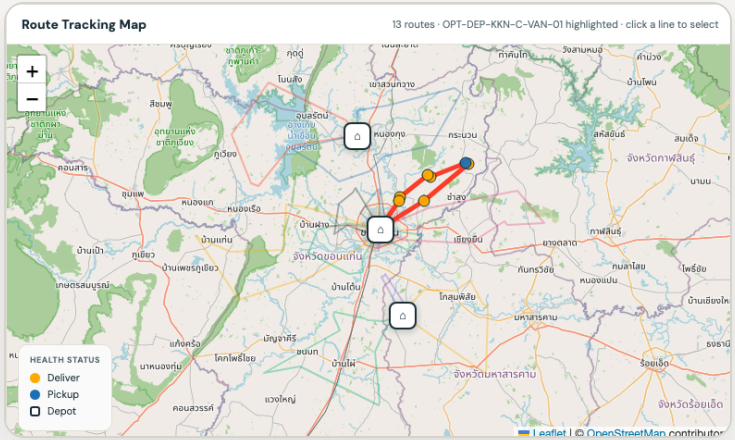
PLAN · FORECAST + OPTIMIZER

CIT COST / DAY	DISTANCE	VANS USED	IDLE CASH	CASH-OUT RISK NODES	SLA	CREW HOURS
฿70K	1,323 km	13 / 30	฿1518.30M	0	100.0%	62 h
▼ 7.8% vs baseline	▼ 22.7% vs baseline	▲ 18.2% vs baseline	▼ 2.8% vs baseline	▼ 100.0% vs baseline	▲ 580.3% vs baseline	▼ 33.2% vs baseline

TODAY · 11 ROUTES, 135 STOPS



PLAN · 13 ROUTES, 97 STOPS



CIT cost/day ฿75K → ฿70K

Distance 1,711 → 1,323 km

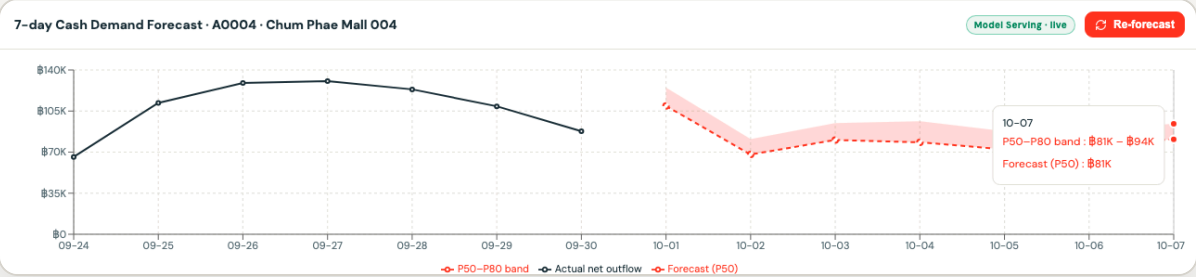
Cash-out risk 64 → 0

SLA 14.7% → 100%

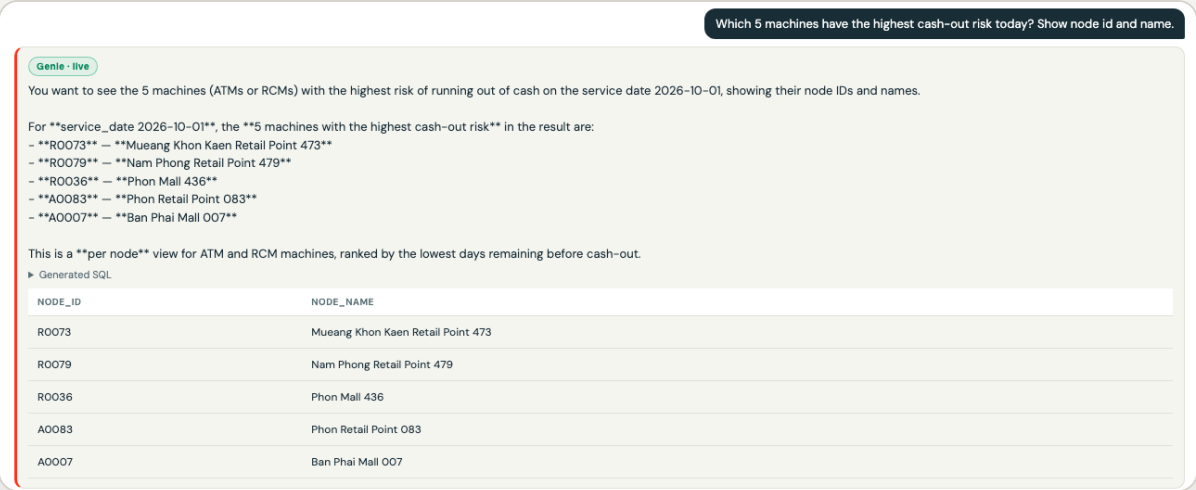
BACKUP · LIVE APP SCREENSHOTS

Backup: forecast, override, ask

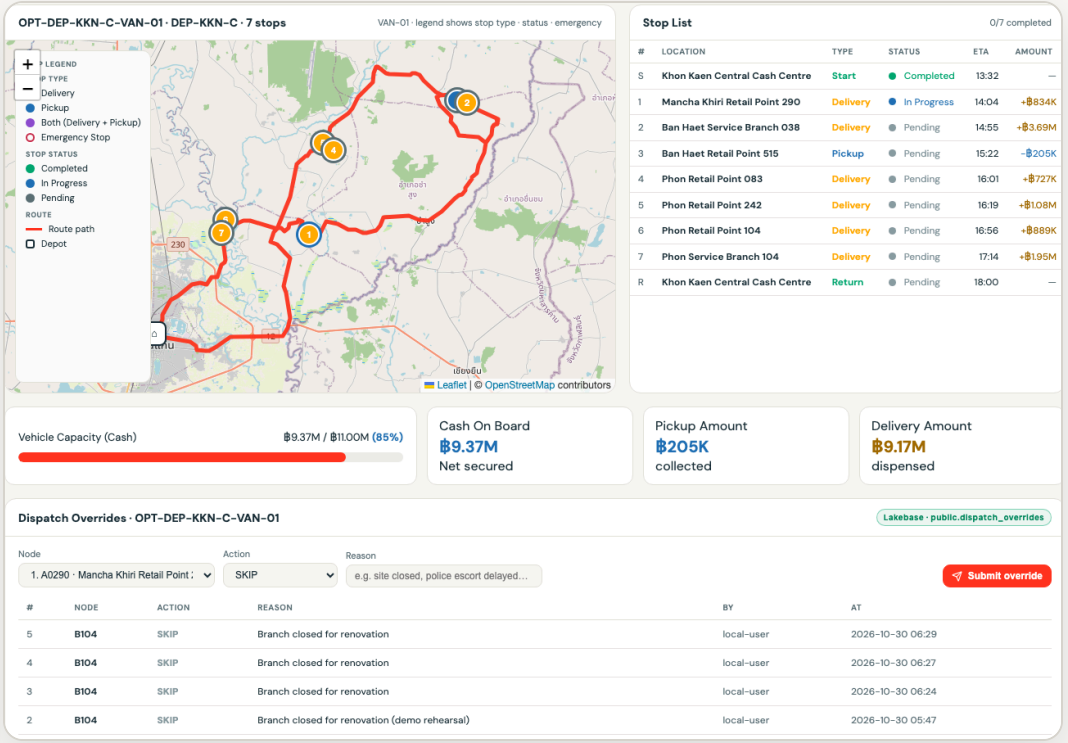
MACHINES · 7-DAY RE-FORECAST FROM MODEL SERVING



ASK CASH OPS · GENIE



TRUCK DETAIL · OVERRIDE SAVED TO LAKEBASE



Every override carries an action, a reason, an author and a timestamp.

SAME ESTATE, SAME DAY · FIXED CYCLE VS FORECAST + OPTIMIZER

The outcome

CIT COST / DAY

THB 75,400 →

THB 69,550

▼ 8% · ~THB 1.5M a year

CASH-OUT RISK

64 machines →

0

Every at-risk machine visited in time

ON-TIME SLA

14.7% →

100%

All stops inside opening hours

ROUTE KM

1,711 km →

1,323 km

▼ 23%

IDLE CASH

released on the plan day

THB 43M

Non-earning float freed

FORECAST ERROR (WAPE)

39% naive →

25%

Better signal, smaller safety stock

Honest note: the plan uses **13 vans instead of 11**: two extra crews to serve real demand on time. Cost still falls because **38 needless stops** are removed.

FROM MVP TO PILOT

Production readiness

READY

- ✓ Defined as code in one Databricks Asset Bundle
- ✓ Serverless pipeline, jobs, SQL, serving, Lakebase
- ✓ App runs as a service principal with scoped access
- ✓ Data-quality expectations on every feed
- ✓ Model versioned in Unity Catalog
- ✓ Audited override writes in Lakebase

GAPS TO CLOSE IN THE PILOT

- ○ IdP groups behind the row filters
- ○ Maker-checker on overrides
- ○ CI/CD with dev and prod targets
- ○ Model drift monitoring
- ○ Road-network travel times
- ○ Full end-to-end run on real feeds

PATH

1. **30-day pilot**
   Khon Kaen Central cash centre
2. **3 cash centres**
   Whole Khon Kaen region
3. **Harden**
   Close the gaps; security and PDPA sign-off
4. **Extend**
   Denominations, vaults, other regions

RISKS

Delivery risks and how we handle them

RISK	HOW WE HANDLE IT
Firewall / on-prem ATM switch	Start with secure file drops into a Unity Catalog volume, then move to private connectivity. A Databricks networking specialist joins.
Timeline slips	Protect the outcome and cut scope: one depot, file feeds first.
PDPA / security sign-off	Data stays in your workspace and region, PII is masked, every change has an audit trail. Our security team joins your review.
Commercials	Owned by the account team; no commitments today.

THE ASK AND NEXT STEPS

Pilot one cash centre for 30 CIT days

≥ 20%
lower CIT cost or km

No higher
cash-outs than today

≥ 80%
of recommended stops accepted

NEXT STEPS

Business sponsor
2-hour scoping session next week with the Cash Ops lead

IT & Security
Agree data feeds, network path and access model

Databricks
Account team for commercials + specialist for networking

Repo: https://github.com/nan-databricks/fe-bar-cash-route

APPENDIX

Appendix: decisions and trade-offs

DECISION	WHY	TRADE-OFF
Declarative pipeline, not notebooks	Expectations, lineage, incremental by default	Less ad-hoc flexibility
One global LightGBM, not 679 models	Shares signal, handles new ATMs, one endpoint	Less per-node tuning
P80 quantile for replenishment	A cash-out costs more than a little extra cash	Some extra float carried
OR-Tools, 25 s per depot	Open source, explainable, fast	Not provably optimal
Lakebase synced tables for the App	Fast reads plus writable overrides	Snapshot freshness, not streaming
Genie agent mode, not a custom agent	Governed by Unity Catalog, no prompt plumbing, fast to value	Less custom tool control
Serverless everywhere	No cluster ops, scale to zero for a once-a-day plan	Cold starts

APPENDIX

Appendix: under the hood

LAKEFLOW

- **Auto Loader** ingests **8 feeds** from a Unity Catalog volume
- **Expectations:** drop null keys and negative cash; warn on bad coordinates
- **Bronze** → **silver** → **gold**; event log and quality metrics per update

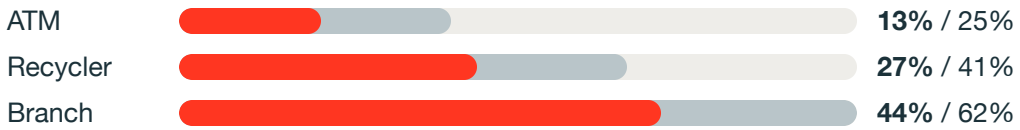
UNITY CATALOG

- **Column mask** on custodian name and phone
- **Row filter** by depot: each cash centre sees its own trips
- **Tags** for layer, domain, PII · **lineage** raw file → app

ML

- **One global LightGBM**: point forecast + P80 quantile, tracked in MLflow
- `cash_demand_forecast@champion` in UC, on a serving endpoint

HOLDOUT WAPE · MODEL / NAIVE



OPTIMIZER

- **CVRP-TW per depot** with OR-Tools: van cash capacity, opening hours, 07:00–18:00 shift
- **97 orders:** 75 critical deliveries, 4 normal deliveries, 18 pickups of surplus cash
- Fixed-cycle baseline written in the same schema for like-for-like KPIs